Various Tools Needed For T&M OF DISPLAYS

For a fully-automated quality control of displays, systems can be implemented throughout the manufacturing process. Makers of smartphones, tablets and other display devices rely on testing to ensure a perfect end-user experience

EFY BUREAU

Devices using displays must be tested and characterised by original equipment manufacturers (OEMs), integrators and those who purchase or use them in bulk. This should be done at R&D, quality control and production line levels to ensure reliability and longer life span of the displays. Applications that require a large number of equal performing units also require test and characterisation tools. The measurement system must also be configured in such a manner that it does not introduce errors into the process.

Fig. 1: Near-eye display measurement system (Credit: www.gamma-sci.com)

Irrespective of LCD, LED, OLED or QLED displays, parameters such as chromaticity, gamma, white balance, contrast, flicker, luminance uniformity and colour uniformity are important to characterise. For displays, evaluation starts with the highest image and signal quality. For testing LCDs, viewing angle is critical, so a straight-ahead direct view of each display is required. It is recommended to carry out the monitor test in a dark room to precisely assess dark image areas.

There are a lot of variables to test when objectively evaluating performance of a display, including brightness, black levels, colour accuracy and more. To test this, different tools are needed depending on what is being tested.

Display measurement and testing devices

Some of the T&M devices for displays are described below.

Luminance meter. This is used when only measurement of a display's luminous intensity in units of a nit (cd/m^2) or foot-lamberts (fl) is required.

Night vision imaging system (NVIS). For manufacturers of military displays and other lighted instrumentation used in conjunction with NVIS, compliance with MIL-L-85762A and MIL-STD-3009 is required. This covers a range of 360nm to 930nm, which requires a high signal-to-noise measurement ratio of 10:1 or greater.

EIZO monitor test. This enables quick and easy assessment of the monitor's image quality. It enables carrying out 13 individual tests to check uniformity of the image display across the entire monitor. Tests include test